6:02

You
4 minutes ago

MC210202542: TAHIR SHAH

BT503 - Environment Biotechnology (Quiz 4)

Question # 7 of 10 (Start time: 08:44:27 AM, 05 February 2023)

All of the following living being is used as biological test for pollution detection except?

Select the correct option

- Animals
- Plants
- Biosensors
- Microbes

Marked by Muhammad Kashif

6:02

You
4 minutes ago

MC210202542: TANIA SABIR

BT503 - Environment Biotechnology (Quiz 4)

Question # 6 of 10 (Start time: 08:43:13 AM, 05 February 2023)

-------technique is used to immobilize the pollutants.

Select the correct option

- Rhizofiltration
- Bioaccumulation
- All of the above
- Phytoextraction

Marked by Muhammad Kashif

6:02

You
4 minutes ago

BT503 - Environment Biotechnology (Quiz 4) Quiz Start Time: 04:00 PM, 05 February 2023

Question # 3 of 10 (Start time: 04:01:42 PM, 05 February 2023) Total Marks: 1

Metabolites produced during phytodegradation are ----------in nature.

Select the correct option

- Toxic
- Both
- Non-toxic
- None of the above

Click to Save Answer & Move to Next Question

Marked by Muhammad Kashif

6:01

You
4 minutes ago

MC[illegible]203369: SANA KHALEEL

BT503 - Environment Biotechnology (Quiz 4)

Question # 6 of 10 (Start time: 04:49:19 PM, 05 February 2023)

-------technique is used to lock the pollutants within biomass of living organisms.

Select the correct option

- Rhizofiltration
- Bioaccumulation
- Phytoextraction
- All of the above

Marked by Muhammad Kashif

6:00

BT503 - Environment Biotechnology (Quiz 4)

You
2 minutes ago

Question # 6 o... (...:43:13 AM, 05 February 2023)

-------technique is used to immobilize the pollutants.

Select the correct option

- Rhizofiltration
- Bioaccumulation
- All of the above
- Phytoextraction

Marked by Muhammad Kashif

5:52

Select text

1. Define Halophiles. **2 MARKS**

Halophiles are organisms that live in extremely salty environments. The name 'halophile' means 'salt-loving' in Greek. Halophiles are all microorganisms. Most of them are bacteria, while some are very primitive eukaryotes. **Eukaryotes** are more complex organisms with a nucleus and membrane-bound organelles.

Halophiles are found in salty places, such as the Great Salt Lake in Utah and the Dead Sea. They are unique because they require high levels of salt that would be lethal to most organisms.

A variety of organisms including a number of bacteria and photosynthetic eukaryotes can tolerate these circumstances, often with an optimum functional temperature as low as 4 ◦ C and stopping reproduction above 12 or 15 ◦ C. Intensely saline environments, such as existing natural salt lakes or within the artificial confines of constructed salt evaporation ponds are home to a group of extremophiles, termed the halophiles.

2. Define psychrophiles. **2 MARKS**

The average oceanic temperature is around 1–3 ◦ C and vast areas of the global land mass are permanently or near-permanently frozen. In these seemingly inhospitable conditions, extremophiles, known as psychrophiles, flourish

---

An organism that thrives in cold temperatures, i.e. ranging from −20 °C to +10 °C
A psychrophile is an organism that is capable of living and thriving in temperatures ranging from −20 °C to +10 °C. The term is often used in classifying microorganisms based on their optimal temperature

3. Define Biosensors. **2 MARKS**

A **biosensor** is an analytical device, used for the detection of a chemical substance, that combines a biological component with a physicochemical detector. The *sensitive biological element*, e.g. tissue, microorganisms, organelles, cell receptors, enzymes, antibodies, nucleic acids, etc., is a biologically derived material or biomimetic component that interacts with, binds with, or recognizes the analyte under study.

4. Define co-metabolism. **2 MARKS**

MUHAMMAD IMRAN

Marked by Muhammad Kashif

Co-metabolism is defined as the simultaneous degradation of two compounds, in which the degradation of the second compound (the secondary substrate) depends on the presence of the first compound (the primary substrate)

The contaminant is not used as food by the microbe, being metabolized fortuitously alongside the organism's food into a less hazardous chemical. Typically, there is no apparent benefit to the microorganisms involved.

Swipe up for filters

So not a living organism

against PAHs (polycyclic aromatic hydrocarbons, e.g., naphthalene and phenanthrene). Bacteria isolated from the same environments may vary in their abilities to degrade PAHs indicative of diverse catabolic pathways Polychlorinated biphenyls (PCBs) are xenobiotics having

Ijazz Jazz

5:52

Select text

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5. How microbes are poly cyclic aromatic compounds. 2

Novel bacteria with capacity to degrade toxic, carcinogenic or teratogenic compounds. Bacte against PAHs (polycyclic aromatic hydrocarbons, e.g., naphthalene and phenanthrene). Bacter isolated from the same environments may vary in their abilities to degrade PAHs indicative of diverse catabolic pathways Polychlorinated biphenyls (PCBs) are xenobiotics having high level

5:46

Select text

years, cadmium concentration decreased by 18%.

Phytoextraction:

Hyperaccumulation

Pteris vittata, accumulates 2.3% arsenic, Certain strains alpine pennycress (*Thlaspi caerulescens*) can bioaccumulate around 1.5% cadmium. Hyperaccumulation of metabolically required elements such as copper or zinc, can be viewed as the outcome of an over-efficient natural mechanism.However, uptake of a completely non-essential metal, particularly in such amounts, remains open to speculation…

Plants are chosen…Permitted to grow for a suitable length of time,Are harvested and the metal accumulated is permanently removed from the original site.Composting or incineration…

Dried plant biomass could be taken to processing works for recycling and for metals like gold… By contrast, low value materials, like lead, would not be a feasible prospect… Potential for biomining metal out of former mines where further traditional methods are no longer practical

152. Terrestrial Phyto-Systems Rhizofiltration

Rhizofiltration

Absorption, Adsorption, orPrecipitation, onto plant roots of contaminants present in the soil water

The principle difference between this and the previous approach: rhizofiltration is typically used to deal with contamination in the water, rather than within the soil itself. The plants to be used are normally brought on hydroponically and gradually acclimatized to the specific character of the water which requires to be treated

Plantation on the site, Begin taking up the solution of pollutants, Harvesting, Final treatment…

Phytostabilization

Close similarities with both phytoextraction and rhizofiltration…Unlike the preceding regimes, ha… this sense, it does not remove the pollutants, but immobilizes them, or locks them up within the plant biomass…

153. Terrestrial Phyto-Systems (TSP)

Wide variety of organic pollutants such as pesticides, solvents and lubricants, etc.Tend to adhere closely to soil particles and are localized within 2m of the surface. Since they are effectively in direct contact with the rhizosphere, they are ideal candidates for phytoremediation.

Typical mechanisms of organic phytoremediation are:

MUHAMMAD IMRAN

Marked by Muhammad Kashif

close

lock

94

- Phytodegradation,
- Rhizodegradation and
- Phytovolatilisation.

Phytodegradation

Biological breakdown of contaminants by plants.

Either

Internally, *having first been taken up by the plant, either directly, using enzym…* then

This leads to biodegradation of the parent substances and subsequent incorporation into plant tissues

Swipe up for filters

So *All* option is correct

Rhizodegradation

Also described as phytostimulation or enhanced rhizospheric biodegradation. Refers to

ljazz Jazz

5:37

Select text

The principle difference between this and the previous approach: rhizofiltration is typically used to deal with contamination in the groundwater, rather than within the soil itself. The plants to be used are normally brought on hydroponically and gradually acclimatized to the specific character of the water which requires to be treated.

Plantation on the site, Begin taking up the solution of pollutants, Harvesting, Final treatment…

Phytostabilization

Close similarities with both phytoextraction and rhizofiltration…Unlike the preceding regimes, harvesting the grown plants is not a feature of the process.In this sense, it does not remove the pollutants, but immobilizes them, or locks them up within the plant biomass…

153. Terrestrial Phyto-Systems (TSP)

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Typical mechanisms of organic phytoremediation are:

- Phytodegradation,
- Rhizodegradation and
- Phytovolatilisation.

Phytodegradation

Biological breakdown of contaminants by plants.

Either

Internally, *having first been taken up by the plants*, or externally, *using enzymes secreted by them.*

This leads to biodegradation into simpler substances and subsequent incorporation into the plant tissues…

Marked by Muhammad Kashif

Phytodegradation

Important that the metabolites which accumulate are either non-toxic, or at least significantly less toxic than the original pollutant…

Rhizodegradation

Also described as phytostimulation or enhanced rhizospheric biodegradation. Refers to the biodegradation of contaminants in the soil by edaphic microbes enhanced by the inherent character of the rhizosphere itself. Rhizosphere supports high microbial biomass. Consequently increases the speed and efficiency of the biodegradation of organic substances within the rhizosphere as compared to other soil regions and microfloral communities. Plant roots increase the soil oxygenation in their vicinity and exude metabolites into the rhizosphere… Denitrifying bacteria, *Pseudomonas* spp., and general heterotrophs are the principal beneficiaries.

Mycorrhizae fungi is also there...

Unique enzymatic pathways that enable the biodegradation of organic substances that could not otherwise be transformed solely by bacterial action. A much slower process than the previously described phytodegradation…

Phytovolatilization

Involves the uptake of the contaminants by plants and their release into the atmosphere, typically in a modified form.

Relies on the transpiration pull of fast growing trees, which accelerates the uptake of the pollutants in ground water solution, which are then transformed and released through the leaves.

E.g., genetically modified variety of the Yellow Poplar, *Liriodendron tulipifera* Engineered by the introduction of mercuric reductase gene (*mer A*). This confers the ability to tolerate higher mercury concentrations, and removal of the pollutant from the soil and volatize it.In a number of poplars have been shown to be able to volatilize around 90% of the TCE (Trichloroeth up.

Is any danger from this kind of pollutant release into the atmosphere???

Phytoremediation tends to be limited to sites where the pollutants are located fairly clos surface. However, recent research has shown that deeply penetrating roots of trees allow deeper contamination to be treated.

154. Phytoremediation – Plant Selection

Major criteria for plant selection are

the particular requirements for the method to be employed, and the nature of the contaminants involved.